Design Patterns Refactoring – Parkway Project

Overview

- **Branch:** feature/design-patterns-refactor
- **Commits:** 2 (Backend + Frontend)
- **Lines Added:** 900+ | **Duplicate Code Removed:** ~400 lines

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Patterns Applied

#	Pattern	Location	Purpose	Status
1	Adapter + Facade	web/src/services/apiClient.js	Centralize API calls, remove hardcoded URLs	✅ Complete
2	Factory	backend/.../util/DTOMapper.java	Consolidate duplicate DTO conversions	✅ Complete
3	@ControllerAdvice	backend/.../exception/GlobalExceptionHandler.java	Remove repeated try-catch blocks	✅ Complete
4	Facade	backend/.../service/AdminCreationFacade.java	Single method for admin + slot creation	✅ Complete
5	Custom Hooks	web/src/hooks/*.js	Simplify scattered	✅ Complete

# Pattern	Location	Purpose	Status
		useState logic	

Pattern	Justification (Why)	Improvement (What it brought)	Notes
Adapter + Facade (API Client)	Centralizes API calls, removes hardcoded URLs, supports dev/prod switching	Single source of truth, consistent error handling, easier maintenance	✅ Explicit enough
Factory (DTOMapper)	Consolidates duplicate DTO conversions across services	Removes ~50 lines duplicate code, null-safe, reusable, testable	✅ Good
@ControllerAdvice (Global Exception Handler)	Removes repeated try-catch blocks across controllers	Consistent error responses, centralized logging, cleaner controllers	✅ Good
Facade (AdminCreationFacade)	Avoids repeating admin+slot creation logic in multiple places	Atomic operation, eliminates duplicates, maintainable, clear code	✅ Good
Custom Hooks	Simplifies scattered useState logic in multiple components	Reduces 40+ useState calls, reusable, cleaner component code, easier testing	✅ Good

Key Improvements

1. Frontend – API Client Service

Problem: Components used 12+ hardcoded API endpoints, inconsistent error handling.

Solution: apiClient.js centralizes all API calls and supports environment switching (.env.local).

Before:

```
fetch('http://localhost:8080/api/users/login', {...})
fetch(`http://localhost:8080/api/bookings/user/${userId}`)
```

After:

```
import { authAPI, bookingAPI } from './services/apiClient';
const user = await authAPI.login(email, password);
const bookings = await bookingAPI.getUserBookings(userId);
```

Benefits:

- Single source of truth for URLs
- Easy dev/prod switching
- Consistent error handling

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2. Backend – DTO Factory

Problem: Duplicate conversion logic in 5+ services.

Solution: DTOMapper centralizes all entity → DTO conversions.

Before: Each service manually mapped fields.

After:

```
@Autowired
private DTOMapper dtoMapper;

List<BookingDTO> bookings = bookingRepository.findByUser(userId)
  .stream()
  .map(dtoMapper::toBookingDTO)
  .collect(Collectors.toList());
```

Benefits:

- Removes ~50 lines of duplicate code
- Null-safe, reusable, testable
- Easier maintenance

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3. Backend – Global Exception Handler

Problem: 50+ repeated try-catch blocks with inconsistent responses.

Solution: @RestControllerAdvice handles exceptions globally.

Before: Each controller manually caught exceptions.

After: Controllers are clean; exceptions handled in one place.

Benefits:

- Consistent error responses
- Centralized logging
- Simplifies controller code

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4. Backend – Admin Creation Facade

Problem: Admin registration repeated slot creation logic in 3 places.

Solution: Facade provides one atomic method to register admin + slots.

Benefits:

- Eliminates duplicate code
- Atomic operation with @Transactional
- Clear, maintainable code

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5. Frontend – Custom Hooks

Problem: 12–15 useState calls scattered across components.

Solution: Create reusable hooks: useModalState, useFormHandler, useCurrentUser.

Benefits:

- Reduces 40+ useState calls to a few hooks
- Reusable, testable
- Cleaner component code

Integration Checklist

Backend:

- Inject DTOMapper where needed
- Replace manual conversions with dtoMapper.to*DTO()
- Remove try-catch in controllers (rely on GlobalExceptionHandler)
- Use AdminCreationFacade for admin registration

Frontend:

- Create .env.local
- Replace direct fetch calls with apiClient methods
- Replace scattered useState with custom hooks
- Test API calls and environment switching

Testing:

- Backend: mvn test
- Frontend: npm test
- Manual: simulate API errors, verify state handling